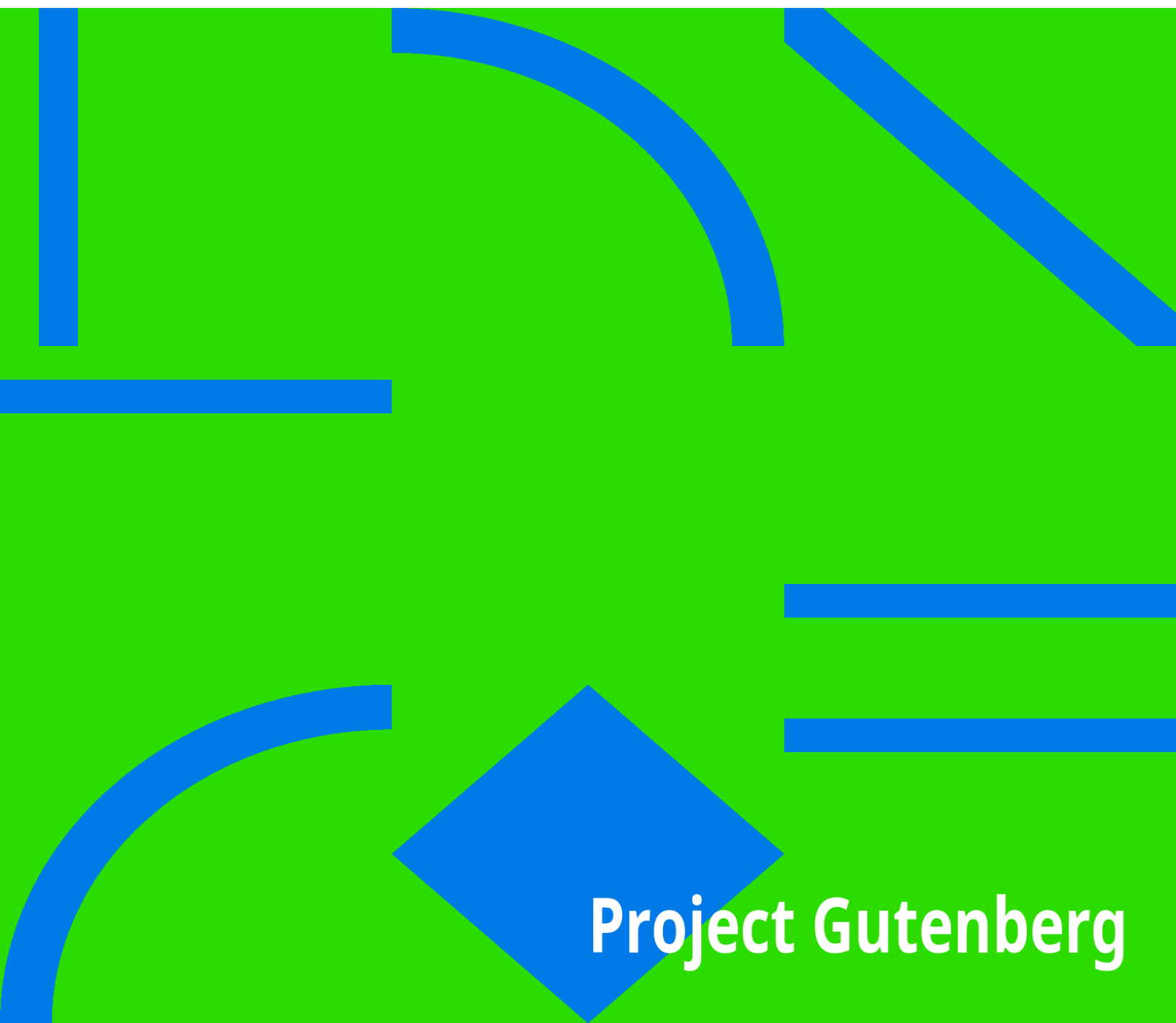


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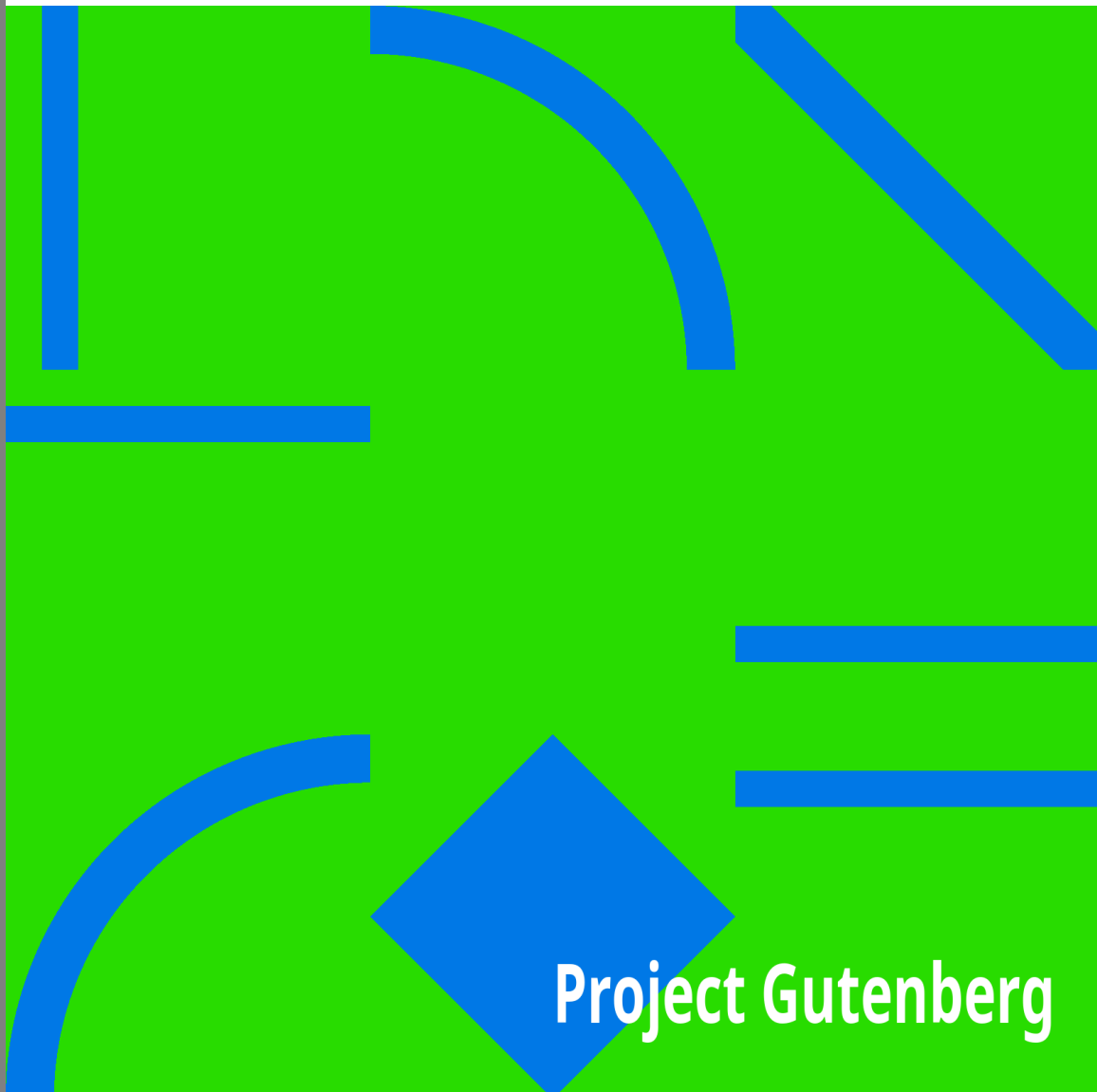
Randall Garrett



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Time Fuze

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*** START OF THE PROJECT GUTENBERG EBOOK TIME FUZE ***



The ultradrive had just one slight drawback: it set up a shock wave that made suns explode. Which made the problem of getting back home a delicate one indeed....

Time Fuze

By Randall Garrett

Illustrated by Paul Orban



Commander Benedict kept his eyes on the rear plate as he activated the intercom. "All right, cut the power. We ought to be safe enough here."

As he released the intercom, Dr. Leicher, of the astronomical staff, stepped up to his side. "Perfectly safe," he nodded, "although even at this distance a star going nova ought to be quite a display."

Benedict didn't shift his gaze from the plate. "Do you have your instruments set up?"

"Not quite. But we have plenty of time. The light won't reach us for several hours yet. Remember, we were outracing it at ten lights."

The commander finally turned, slowly letting his breath out in a soft sigh. "Dr. Leicher, I would say that this is just about the foulest coincidence that could happen to the first interstellar vessel ever to leave the Solar System."

Leicher shrugged. "In one way of thinking, yes. It is certainly true that we will never know, now, whether Alpha Centauri A ever had any planets. But, in another way, it is extremely fortunate that we should be so near a stellar explosion because of the wealth of scientific information we can obtain. As you say, it is a

coincidence, and probably one that happens only once in a billion years. The chances of any particular star going nova are small. That we should be so close when it happens is of a vanishingly small order of probability."

Commander Benedict took off his cap and looked at the damp stain in the sweatband. "Nevertheless, Doctor, it is damned unnerving to come out of ultradrive a couple of hundred million miles from the first star ever visited by man and have to turn tail and run because the damned thing practically blows up in your face."

Leicher could see that Benedict was upset; he rarely used the same profanity twice in one sentence.

They had been downright lucky, at that. If Leicher hadn't seen the star begin to swell and brighten, if he hadn't known what it meant, or if Commander Benedict hadn't been quick enough in shifting the ship back into ultradrive—Leicher had a vision of an incandescent cloud of gaseous metal that had once been a spaceship.

The intercom buzzed. The commander answered, "Yes?"

"Sir, would you tell Dr. Leicher that we have everything set up now?"

Leicher nodded and turned to leave. "I guess we have nothing to do now but wait."

When the light from the nova did come, Commander Benedict was back at the plate again—the forward one, this time, since the ship had been turned around in order to align the astronomy lab in the nose with the star.

Alpha Centauri A began to brighten and spread. It made Benedict think of a light bulb connected through a rheostat, with someone turning that rheostat, turning it until the circuit was well overloaded.

The light began to hurt Benedict's eyes even at that distance and he had to cut down the receptivity in order to watch. After a while, he turned away from the plate. Not because the show was over, but simply because it had slowed to a point beyond which no change seemed to take place to the human eye.

Five weeks later, much to Leicher's chagrin, Commander Benedict announced that they had to leave the vicinity. The ship had only been provisioned to go to Alpha Centauri, scout the system without landing on any of the planets, and return. At ten lights, top speed for the ultradrive, it would take better than three months to get back.

"I know you'd like to watch it go through the complete cycle," Benedict said, "but we can't go back home as a bunch of starved skeletons."

Leicher resigned himself to the necessity of leaving much of his work unfinished, and, although he knew it was a case of sour grapes, consoled himself with the thought that he could at least get most of the remaining information from the five-hundred-inch telescope on Luna, four years from then.

As the ship slipped into the not-quite-space through which the ultradrive propelled it, Leicher began to consolidate the material he had already gathered.

Commander Benedict wrote in the log:

Fifty-four days out from Sol. Alpha Centauri has long since faded back into its pre-blowup state, since we have far outdistanced the light from its explosion. It now looks as it did two years ago. It—

"Pardon me, Commander," Leicher interrupted, "But I have something interesting to show you."

Benedict took his fingers off the keys and turned around in his chair. "What is it, Doctor?"

Leicher frowned at the papers in his hands. "I've been doing some work on the probability of that explosion happening just as it did, and I've come up with some rather frightening figures. As I said before, the probability was small. A little calculation has given us some information which makes it even smaller. For instance: with a possible error of plus or minus two

seconds Alpha Centauri A began to explode the instant we came out of ultradrive!

"Now, the probability of that occurring comes out so small that it should happen only once in ten to the four hundred sixty-seventh seconds."

It was Commander Benedict's turn to frown. "So?"

"Commander, the entire universe is only about ten to the seventeenth seconds old. But to give you an idea, let's say that the chances of its happening are *once* in millions of trillions of years!"

Benedict blinked. The number, he realized, was totally beyond his comprehension—or anyone else's.

"Well, so what? Now it has happened that one time. That simply means that it will almost certainly never happen again!"

"True. But, Commander, when you buck odds like that and win, the thing to do is look for some factor that is cheating in your favor. If you took a pair of dice and started throwing sevens, one right after another—*for the next couple of thousand years*—you'd begin to suspect they were loaded."

Benedict said nothing; he just waited expectantly.

"There is only one thing that could have done it. Our ship." Leicher said it quietly, without emphasis.

"What we know about the hyperspace, or superspace, or whatever it is we move through in ultradrive is almost nothing. Coming out of it so near to a star might set up some sort of shock wave in normal space which would completely disrupt that star's internal balance, resulting in the liberation of unimaginably vast amounts of energy, causing that star to go nova. We can only assume that we ourselves were the fuze that set off that nova."

Benedict stood up slowly. When he spoke, his voice was a choking whisper. "You mean the sun—Sol—might...."

Leicher nodded. "I don't say that it definitely would. But the probability is that we were the cause of the destruction of Alpha Centauri A, and therefore might cause the destruction of Sol in the same way."

Benedict's voice was steady again. "That means that we can't go back again, doesn't it? Even if we're not positive, we can't take the chance."

"Not necessarily. We can get fairly close before we cut out the drive, and come in the rest of the way at sub-light speed. It'll take longer, and we'll have to go on half or one-third rations, but we *can* do it!"

"How far away?"

"I don't know what the minimum distance is, but I do know how we can gage a distance. Remember, neither Alpha Centauri B or C were detonated. We'll have to cut our drive at least as far away from Sol as they are from A."

"I see." The commander was silent for a moment, then: "Very well, Dr. Leicher. If that's the safest way, that's the only way."

Benedict issued the orders, while Leicher figured the exact point at which they must cut out the drive, and how long the trip would take. The rations would have to be cut down accordingly.

Commander Benedict's mind whirled around the monstrosity of the whole thing like some dizzy bee around a flower. What if there had been planets around Centauri A? What if they had been inhabited? Had he, all unwittingly, killed entire races of living, intelligent beings?

But, how could he have known? The drive had never been tested before. It couldn't be tested inside the Solar System—it was too fast. He and his crew had been volunteers, knowing that they might die when the drive went on.

Suddenly, Benedict gasped and slammed his fist down on the desk before him.

Leicher looked up. "What's the matter, Commander?"

"Suppose," came the answer, "Just suppose, that we have the same effect on a star when we *go into* ultradrive as we do when we come out of it?"

Leicher was silent for a moment, stunned by the possibility. There was nothing to say, anyway. They could only wait....

A little more than half a light year from Sol, when the ship reached the point where its occupants could see the light that had left their home sun more than seven months before, they watched it become suddenly, horribly brighter. *A hundred thousand times brighter!*

... THE END

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